Assignment

(PR01)

Presented to:
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CR431E : Operating System Management Scripting with PowerShell
Department of Computer Engineering and Software Engineering

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1 Introduction

This semester assignment consists of building a PowerShell module dedicated to firewall log management. Firewall logs are normally long text files that are hard to read and even harder to analyze quickly. The objective of this module is to simplify this process by converting raw log data into structured and readable information.

2 Description of the Use Cases

Our module can be used for several cases. For example :

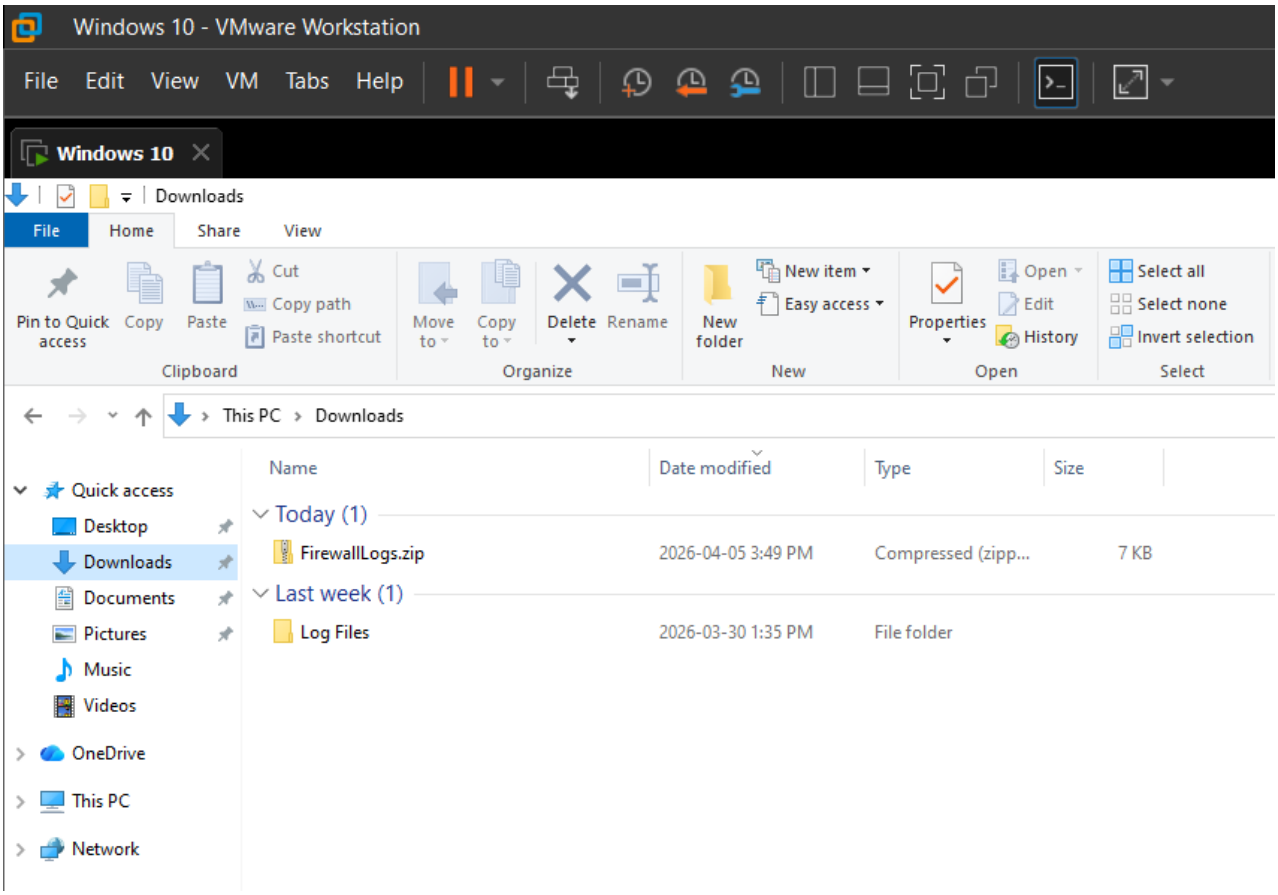
- Log analysis
- DNS resolution

3 Module Installation

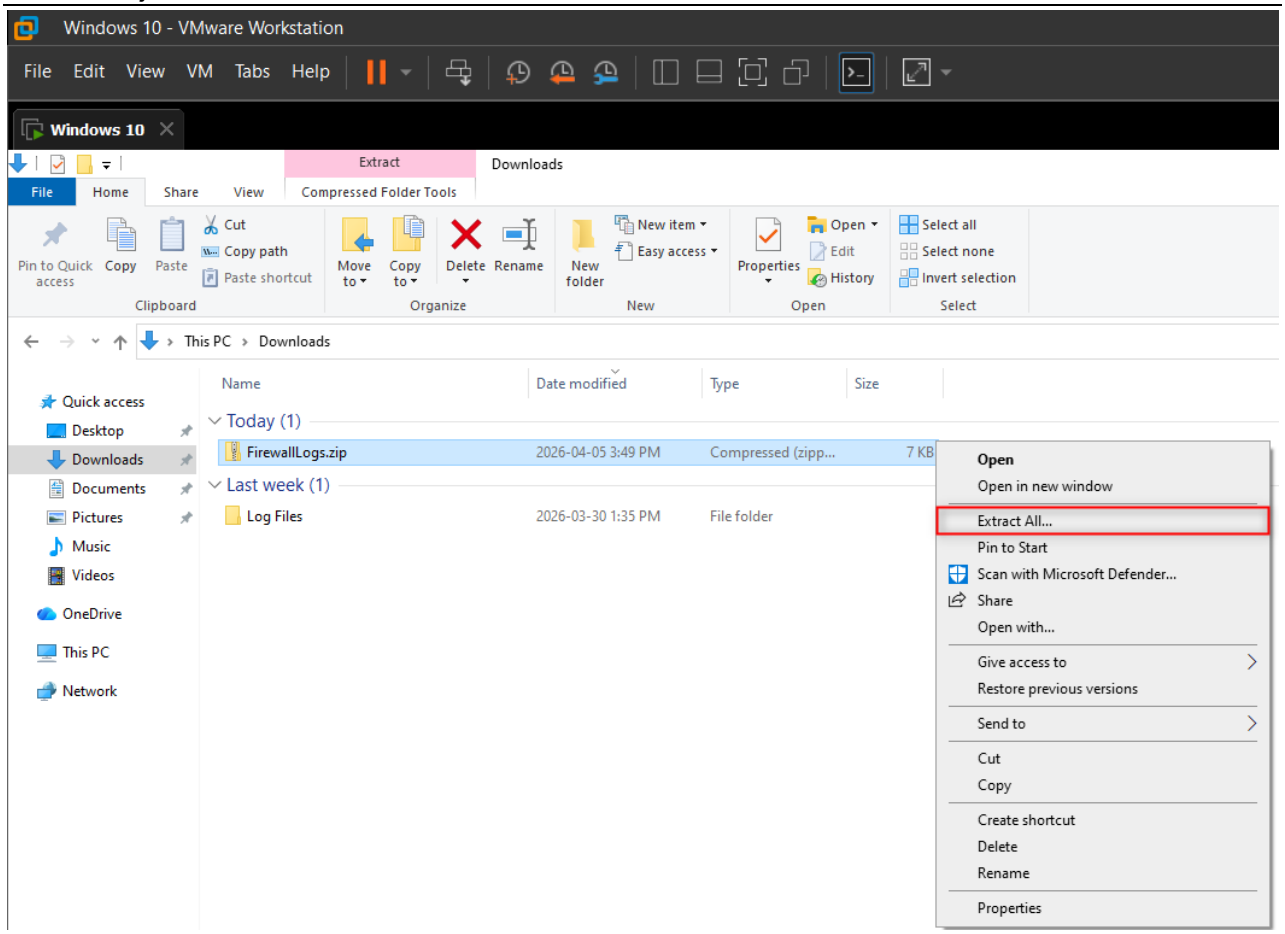
To use our module, you will need to follow these steps.

1. Download the provided ZIP file, which contains our PowerShell module file (FirewallLogs.psm1) and our manifest file (FirewallLogs.psd1). If you haven't already, don't forget to have the required log files for this assignment (fw.log, fw.log.1, pihole.log, pihole.log.1 and pihole.log.2) in your system.

   *We suggest putting them in a folder. For example, LogFiles.



2. Extract the ZIP file.


Windows 10 - VMware Workstation
File Edit View VM Tabs Help
Windows 10
Extract
Downloads
File Home Share View
Compressed Folder Tools
Pin to Quick access
Copy
Paste
Cut
Copy path
Paste shortcut
Clipboard
Move to
Copy to
Delete
Rename
Organize
New folder
New item
Easy access
New
Properties
Open
Edit
History
Open
Select all
Select none
Invert selection
Select
This PC › Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
Today (1)
FirewallLogs.zip
2026-04-05 3:49 PM
Compressed (zipp...
7 KB
Last week (1)
Log Files
2026-03-30 1:35 PM
File folder
Open
Open in new window
Extract All...
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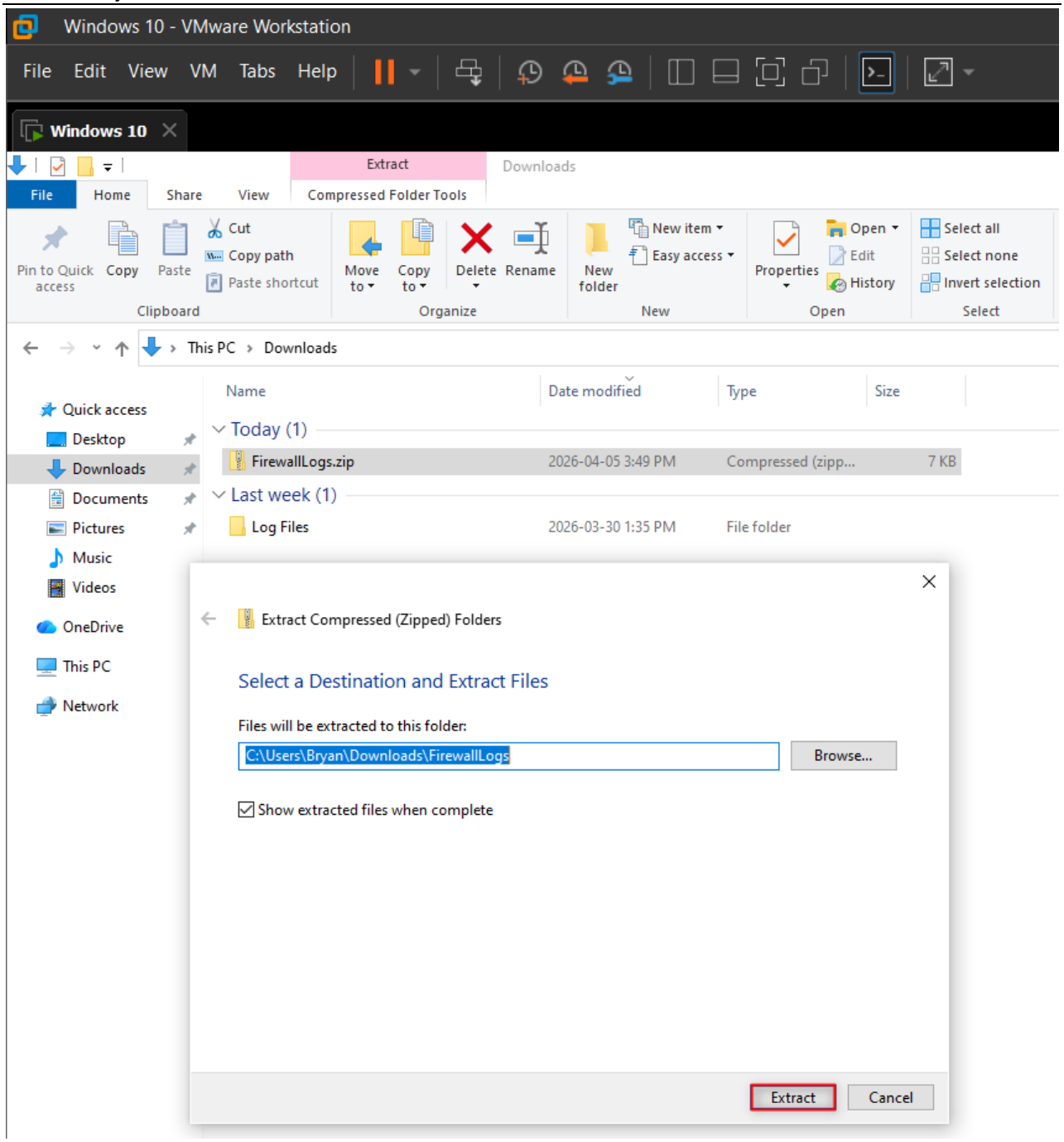
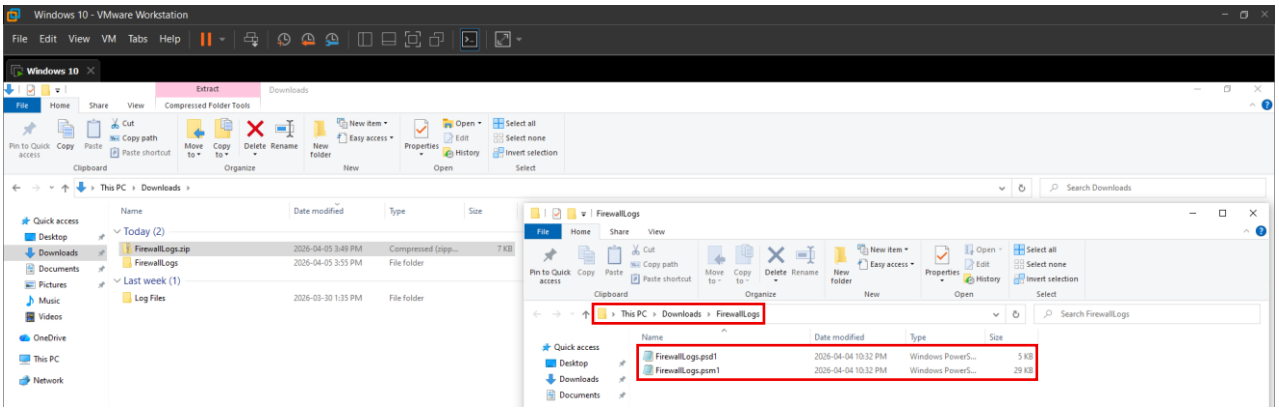
Restore previous versions
Send to
Cut
Copy
Create shortcut
Delete
Rename
Properties





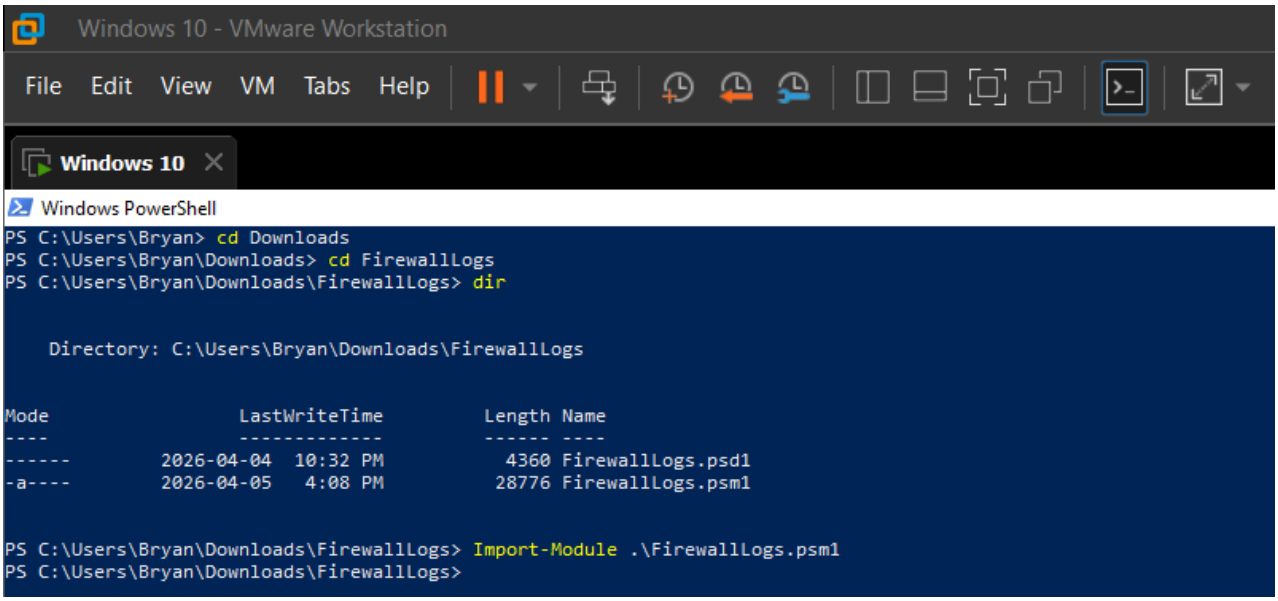
3. Open a PowerShell terminal and navigate to the following folder.

```
cd \Downloads\FirewallLogs
```

In the screenshot of step 4, Bryan accessed the directory one folder at a time (Downloads -> FirewallLogs). He also executed the **dir** command to display the list of files in the newly generated FirewallLogs folder.

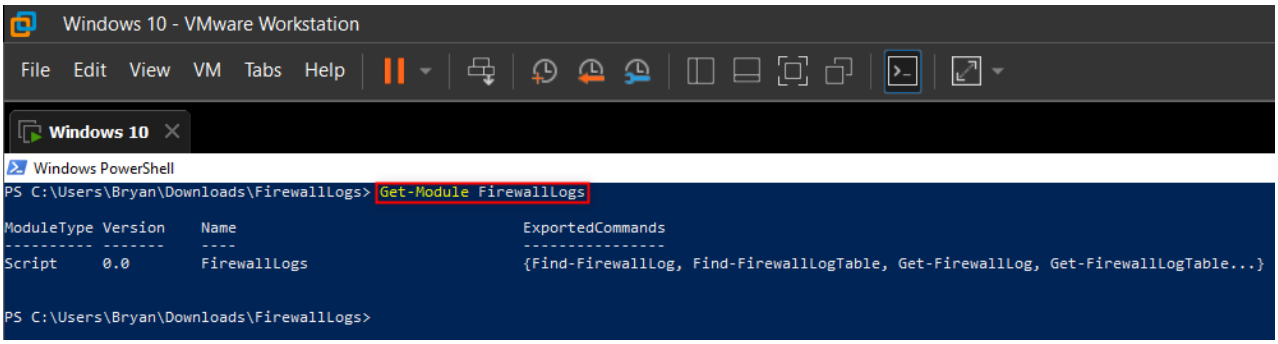
4. Import our FirewallLogs module by running the following command :

```
Import-Module .\FirewallLogs.psm1
```



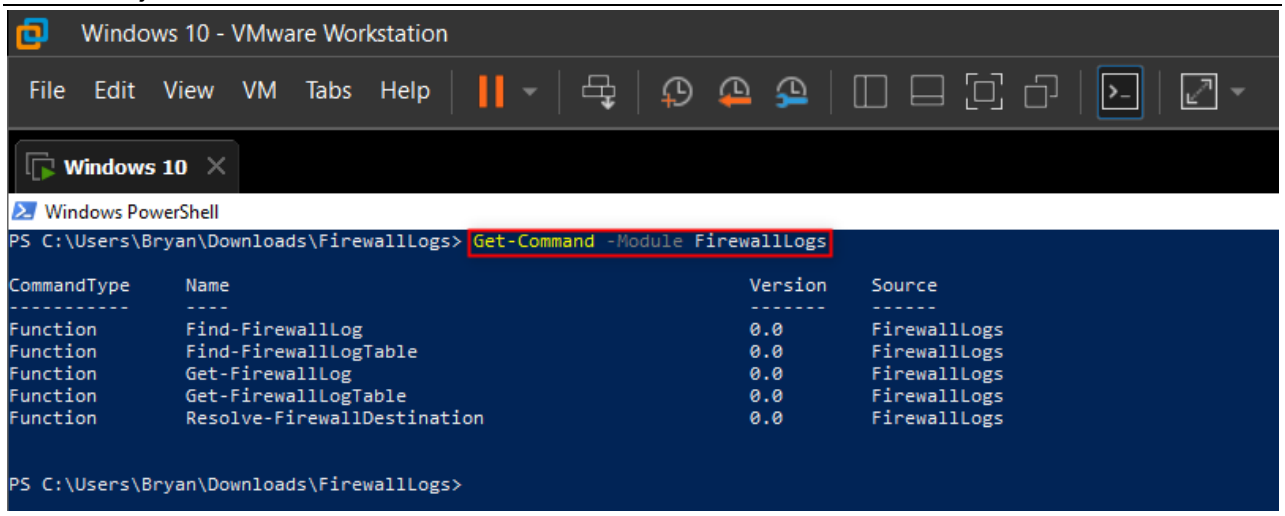
Verify the installation of our module in your system by running the following command :

```
Get-Module FirewallLogs
```



You could also review the list of commands of our FirewallLogs module with this command :

```
Get-Command -Module FirewallLogs
```



4 Dependencies

In addition to the module file and the manifest file, you will need the following dependencies to use this module :

- Windows PowerShell
- Firewall log files (fw.log and fw.log.1)
- Pi-hole log files (pihole.log, pihole.log.1, pihole.log.2)

5 Commands (Functions)

5.1 Function 1: Get-FirewallLog

5.1.1 Usage

The Get-FirewallLog function reads a firewall log file, parses the contents of the log file, and displays the entries in a structured format.

5.1.2 Arguments

This function uses two arguments :

- -FirewallLogPath
  - Alias : -P
- -Full
  - Alias : -F
- Limit
  - Alias : -L

5.1.3 Use of arguments

The FirewallLogPath argument is used to specify the path to a firewall log file. The Full argument is optional. If you specify this argument during your command exécution, it displays all available content fields

from the specified log file. The Limit argument allows you to control how many entries are displayed. By default, all entries are displayed if you do not use argument.

5.1.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path Example : C:\Users\bryan\Downloads\Log Files\fw.log
Full	Type : Switch *There is no format.
Limit	Type : Integer (int) Format : Number Examples : 5, 10, 15, 20, etc.

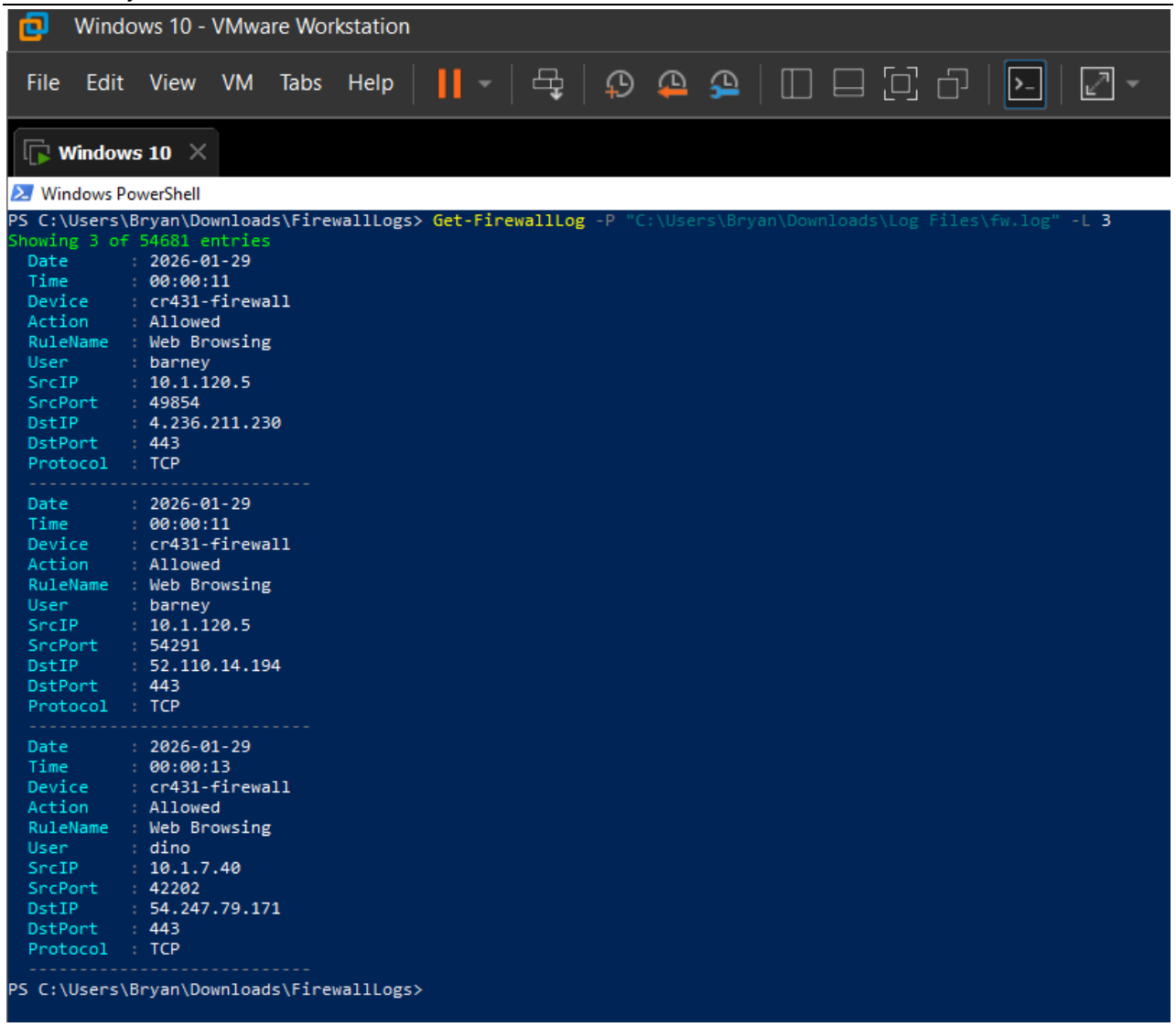
5.1.5 Expected result

Firewall log entries are displayed as structured PowerShell objects. If you only use the FirewallLogPath argument in the Get-FirewallLog command, you will only see the minimal fields (User, Rule name, Source IP address, Destination IP address, etc.).

5.1.6 Examples

Example 1 : Get-FirewallLog -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -L 3

*If you set your limit to a high number (e.g. 10), scroll down your session window to see the rest of your displayed results.



Example 2 : Get-FirewallLog -P "C:\Users\maher\Downloads\FirewallLogs\fw.log.1"

5.2 Function 2: Get-FirewallLogTable

5.2.1 Usage

The Get-FirewallLogTable function reads a firewall log file, parses the contents of the log file, and displays the entries in a structured **table** format. Results can be displayed directly in your Windows PowerShell session or in a GridView window.

5.2.2 Arguments

This function uses two arguments :

- -FirewallLogPath
  - Alias : -P
- -Limit
  - Alias : -L

- -GridView
    - Alias : -G

5.2.3 Use of arguments

The FirewallLogPath argument is used to specify the path to a firewall log file. The Limit argument allows you to control how many entries are displayed. By default, the function shows 200 entries. The GridView argument changes the format of the output results by displaying them in a GridView window.

5.2.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path Example : C:\Users\bryan\Downloads\Log Files\fw.log
Limit	Type : Integer (int) Format : Number (up to 200) Examples : 10, 20, 30, 40, etc.
GridView	Type : Switch

5.2.5 Expected result

The firewall log entries are displayed in a structured table containing minimal information, such as :

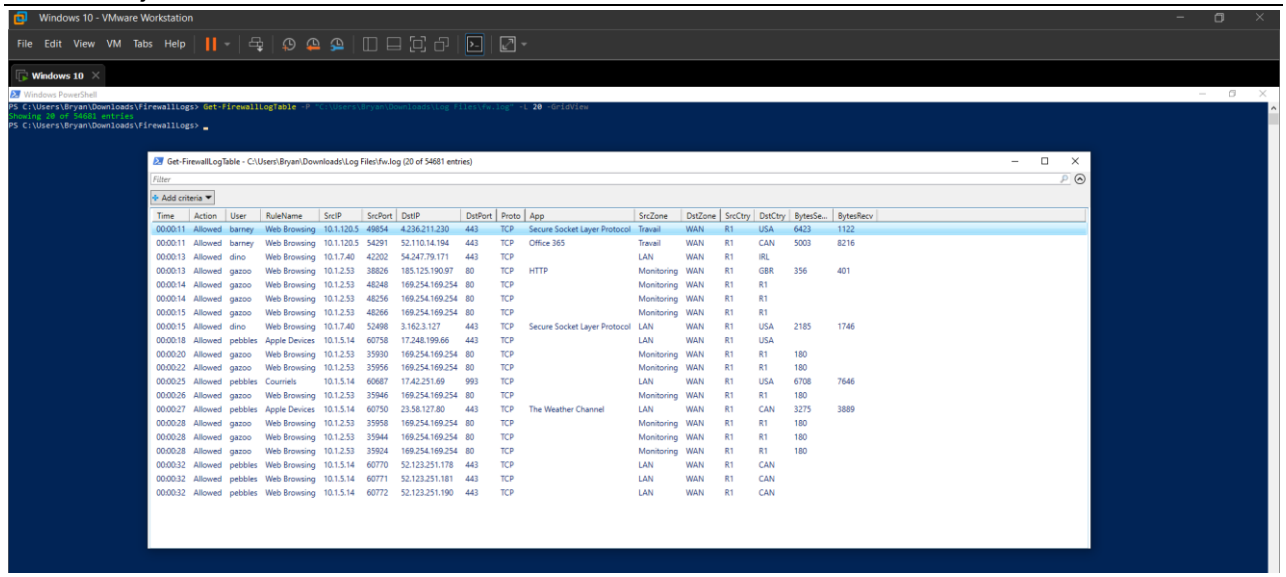
- Time
- Action
- The name of the user
- The name of the rule
- Source IP address
- Source port
- Destination IP address
- Destination port
- Etc…

Rows can also have custom colour depending on the firewall action of the log entry.

5.2.6 Examples

Example 1 : Get-FirewallLogTable -P "C:\Users\maher\Downloads\FirewallLogs\fw.log" -L 50

Example 2 : Get-FirewallLogTable -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -L 20 -GridView



Time	Action	User	RuleName	SrcIP	SrcPort	DstIP	DstPort	Proto	App	SrcZone	DstZone	SrcCtry	DstCtry	BytesSe...	BytesRecv
00:00:11	Allowed	barney	Web Browsing	10.1.120.5	49854	4.236.211.230	443	TCP	Secure Socket Layer Protocol	Travail	WAN	R1	USA	6423	1122
00:00:11	Allowed	barney	Web Browsing	10.1.120.5	54291	52.110.14.194	443	TCP	Office 365	Travail	WAN	R1	CAN	5003	8216
00:00:13	Allowed	dino	Web Browsing	10.1.7.40	42202	54.247.79.171	443	TCP		LAN	WAN	R1	IRL		
00:00:13	Allowed	gazoo	Web Browsing	10.1.2.53	38826	185.125.190.97	80	TCP	HTTP	Monitoring	WAN	R1	GBR	356	401
00:00:14	Allowed	gazoo	Web Browsing	10.1.2.53	48248	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1		
00:00:14	Allowed	gazoo	Web Browsing	10.1.2.53	48256	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1		
00:00:15	Allowed	gazoo	Web Browsing	10.1.2.53	48266	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1		
00:00:15	Allowed	dino	Web Browsing	10.1.7.40	52498	3.162.3.127	443	TCP	Secure Socket Layer Protocol	LAN	WAN	R1	USA	2185	1746
00:00:18	Allowed	pebbles	Apple Devices	10.1.5.14	60758	17.248.199.66	443	TCP		LAN	WAN	R1	USA		
00:00:20	Allowed	gazoo	Web Browsing	10.1.2.53	35930	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:22	Allowed	gazoo	Web Browsing	10.1.2.53	35956	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:25	Allowed	pebbles	Courriels	10.1.5.14	60687	17.42.251.69	993	TCP		LAN	WAN	R1	USA	6708	7646
00:00:26	Allowed	gazoo	Web Browsing	10.1.2.53	35946	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:27	Allowed	pebbles	Apple Devices	10.1.5.14	60750	23.58.127.80	443	TCP	The Weather Channel	LAN	WAN	R1	CAN	3275	3889
00:00:28	Allowed	gazoo	Web Browsing	10.1.2.53	35958	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:28	Allowed	gazoo	Web Browsing	10.1.2.53	35944	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:28	Allowed	gazoo	Web Browsing	10.1.2.53	35924	169.254.169.254	80	TCP		Monitoring	WAN	R1	R1	180	
00:00:32	Allowed	pebbles	Web Browsing	10.1.5.14	60770	52.123.251.178	443	TCP		LAN	WAN	R1	CAN		
00:00:32	Allowed	pebbles	Web Browsing	10.1.5.14	60771	52.123.251.181	443	TCP		LAN	WAN	R1	CAN		
00:00:32	Allowed	pebbles	Web Browsing	10.1.5.14	60772	52.123.251.190	443	TCP		LAN	WAN	R1	CAN		

5.3 Function 3 : Find-FirewallLog

5.3.1 Usage

The Find-FirewallLog function is used to search, filter and display firewall log entries based on five specific filters :

- The source IP address
- The destination IP address
- The destination port number
- The user
- The name of the rule

5.3.2 Arguments

This function uses the following list of arguments :

- -FirewallLogPath
  - Alias : -P
- -SourceIP
  - Alias : -Src
- -DestinationIP
  - Alias : -Dst
- -DestinationPort
  - Alias : -Port
- -User
  - Alias : -User
- -RuleName
  - Alias : -Rule

5.3.3 Use of arguments

The FirewallLogPath argument tells the Find-FirewallLog function which firewall log file (with their path) to read. The other arguments are the five specified filters.

Name of the filter	The role of the filter
SourceIP	Filter the specified log file by source IP address
DestinationIP	Filter the specified log file by destination IP address
DestinationPort	Filter the specified log file by destination port number
User	Filter the specified log file by the name of the user
RuleName	Filter the specified log file by the name of the firewall rule

5.3.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
SourceIP	Type : String Format : IPv4 address
DestinationIP	Type : String Format : IPv4 address
DestinationPort	Type : Integer (int) Format : Port number Example : 443
User	Type : String
RuleName	Type : String

5.3.5 Expected result

Firewall log entries are returned and displayed, only if they match the filters you selected as your argument(s). The number of entry results is also displayed. If there are no matching entries, an error message will appear on your screen.

5.3.6 Examples

Example 1 : Find-FirewallLog -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -SourceIP ""

Example 2 : Find-FirewallLog -P "C:\Users\maher\Downloads\FirewallLogs\fw.log" -DestinationIP "" -DestinationPort ""

5.4 Function 4 : Find-FirewallLogTable

5.4.1 Usage

The Find-FirewallLogTable function is used to search, filters firewall log entries based on the five following filters, and displays the results in a table format :

- The source IP address
- The destination IP address
- The destination port number
- The user
- The name of the rule

5.4.2 Arguments

This function uses the following list of arguments :

- -FirewallLogPath
  - Alias : -P
- -SourceIP
  - Alias : -Src
- -DestinationIP
  - Alias : -Dst
- -DestinationPort
  - Alias : -Port
- -User
  - Alias : -User
- -RuleName
  - Alias : -Rule
- Limit
  - Alias : -L
- GridView
  - Alias : -G

5.4.3 Use of arguments

The FirewallLogPath argument tells the Find-FirewallLog function which firewall log file (with their path) to read. The other arguments are the five specified filters.

Name of the filter	The role of the filter
SourceIP	Filter the specified log file by source IP address
DestinationIP	Filter the specified log file by destination IP address
DestinationPort	Filter the specified log file by destination port number
User	Filter the specified log file by the name of the user
RuleName	Filter the specified log file by the name of the firewall rule

The Limit argument determines the maximum number of log entries that will be displayed.

The GridView argument displays the results in a GridView window in the format of a table.

5.4.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
SourceIP	Type : String Format : IPv4 address
DestinationIP	Type : String Format : IPv4 address
DestinationPort	Type : Integer (int) Format : Port number Example : 443
User	Type : String
RuleName	Type : String
Limit	Type : Integer (int) Format : Number Examples : 10, 20, 30, …, 200
GridView	Type : Switch Output format : Structured table in a GridView window

5.4.5 Expected result

Firewall log entries that match the selected filters are displayed in a structured table format. If matching log entries are found, the function will display the number of results and the list of log entries according to the specified Limit parameter (argument). If no entries match the filters, an error message will inform you that no matching results were found.

5.4.6 Examples

Example 1 : Find-FirewallLogTable -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -SourceIP "" -L

Example 2 : Find-FirewallLogTable -P "C:\Users\maher\Downloads\FirewallLogs\fw.log" -DestinationIP "" -DestinationPort "" -GridView

5.5 Function 5 :Resolve-FirewallDestination

5.5.1 Usage

The Resolve-FirewallDestination function is used to link destination IP addresses from firewall log files to domain names using Pi-hole log files.

5.5.2 Arguments

This function uses two arguments :

- FirewallLogPath
  - Alias : -P
- PiholeLogPath
  - Alias : -DNS

5.5.3 Use of arguments

The FirewallLogFile argument specifies the path of a firewall log file. The PiholeLogFile argument specifies the path to a Pi-hole log file.

5.5.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
PiholeLogPath	Type : String Format : File path

5.5.5 Expected result

The command exécution of the Resolve-FirewallDestination should output the destination IP addresses and their corresponding domain name (if there is a match). If no match is found, the appropriate domain fields will be empty.

5.5.6 Examples

Example 1 : Resolve-FirewallDestination -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -DNS "C:\Users\user\Downloads\FirewallLogs\pihole.log"

Example 2 : Resolve-FirewallDestination -P "C:\Users\user\Downloads\FirewallLogs\fw.log.1" -DNS "C:\Users\user\Downloads\FirewallLogs\pihole.log.2"

6 Conclusion

In conclusion, our firewall log management module, FirewallLogs, makes it much easier to work with firewall logs by converting raw log data into readable log information. It can help you save you a lot of time if you need to filter and investigate network activity by analyzing log files.

7 References

Microsoft Corporation. (s.d.). *cd*. Récupéré sur Microsoft Learn: https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/cd

Microsoft Corporation. (s.d.). *How to Write a PowerShell module manifest*. Récupéré sur Microsoft Learn: https://learn.microsoft.com/en-us/powershell/scripting/developer/module/how-to-write-a-powershell-module-manifest?view=powershell-7.6

Microsoft Corporation. (s.d.). *How To Write a PowerShell Script Module*. Récupéré sur Microsoft Learn: https://learn.microsoft.com/en-us/powershell/scripting/developer/module/how-to-write-a-powershell-script-module?view=powershell-7.6

Microsoft Corporation. (s.d.). *Import-Module*. Récupéré sur Microsoft Learn: https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.core/import-module?view=powershell-7.6

TechThoughts. (2022, July 12). *PowerShell Modules*. Récupéré sur YouTube: https://youtu.be/tg5bEPsXTQY?si=SLEMKHOYA_0gT0_c